

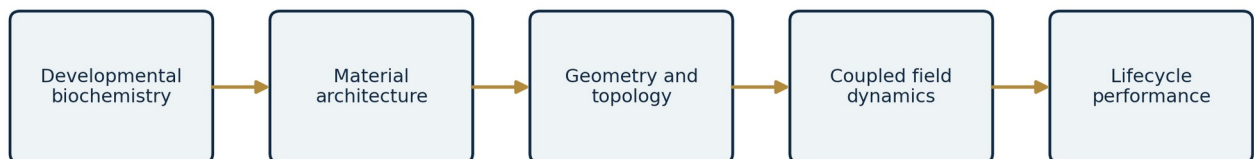
GEOMETRICALLY GOVERNED CHITIN INTERFACES

Resonance, Aerodynamic Coupling, Acoustic Transduction,
and Collective Thermal Regulation in Beetles, Cicadas, and Honeybees

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GEOMETRICALLY GOVERNED CHITIN INTERFACES



Structure, resonance, sensing, flow, sound and heat emerge through one differentiated material continuum.

Evidence-disciplined treatment of established continuum mechanics, measured acoustic transduction, and distributed biological thermoregulation.

Abstract

Arthropod exoskeletons are often described as protective shells, skeletal supports, or attachment structures for muscles. Those descriptions remain correct but incomplete. Chitin-based anatomy also constitutes the spatially organized material interface through which arthropods exchange force, momentum, vibration, sound, heat, and information with their surroundings.

The relevant functional material is not chemically pure chitin. It is a heterogeneous chitin-protein composite whose stiffness, damping, hydration, fiber orientation, thickness, sclerotization, resilin content, geometry, and surface topology vary with anatomical position. Consequently, an insect wing, tymbal, thorax, elytron, abdominal wall, or sensory structure cannot be represented adequately as a uniform object upon which separate functions have later been installed. The material architecture and geometry jointly constitute the operating system.

This paper develops an evidence-disciplined mathematical framework and applies it to three established regimes: aeroelastic flight and surface-mediated sensing in beetles and related insects; mechanical-to-acoustic transduction in singing cicadas; and individual and colony-scale thermal regulation in honeybees, including defensive bee-balling in some Asian honeybee lineages.

biochemistry -> material architecture -> geometry -> coupled field dynamics -> lifecycle performance

The paper does not claim a demonstrated microscopic phonon-assisted lift mechanism or any other exotic source of aerodynamic force. Such possibilities are acknowledged as open subjects requiring their own mathematical definitions and experimental evidence. The present study instead develops what follows from established anatomy, measured behavior, continuum mechanics, fluid-structure interaction, acoustics, heat transfer, and distributed biological control.

1. Central proposition and scope

In arthropods, anatomy and functional material architecture are not separable design layers. Development produces continuous, spatially differentiated chitin-protein composites whose geometry and constitutive properties jointly govern structural, vibrational, aerodynamic, acoustic, sensory, and thermal behavior.

A conventionally engineered system commonly separates frame, joint, actuator, surface, sensor, and controller. Arthropod development instead expresses local differentiation within one continuous anatomy. Regions of that anatomy become springs, resonators, aerodynamic boundaries, radiators, strain sensors, and thermal interfaces through geometry, composition, attachment, and loading.

The structured material is the mechanism.

2. Material architecture and governing equations

2.1 The cuticle as a spatially differentiated composite

Insect cuticle contains chitinous fibrils within a protein matrix. Its properties vary through fibril orientation, lamellar arrangement, hydration, cross-linking, sclerotization, local resilin content, thickness, curvature, corrugation, vein topology, and microtexture. Highly sclerotized regions can reach gigapascal-scale moduli, while resilin-rich regions remain much more compliant. The result is a single material system capable of rigid support, elastic return, joint motion, damping, and damage tolerance.

$$\sigma_{ij} = C_{ijkl}(x) \epsilon_{kl}$$

Here $C_{ijkl}(x)$ denotes a spatially varying anisotropic stiffness tensor. Under harmonic loading a complex constitutive field can be written

$$C^*(x, \omega) = C'(x, \omega) + i C''(x, \omega)$$

$$\tan(\delta) = C'' / C'$$

The real component represents stored response and the imaginary component represents dissipation under the adopted convention. The precise scalar loss tangent is meaningful only after selecting a material direction or reduced constitutive component.

2.2 Geometry as part of the operator

Anatomical geometry defines both the body domain and the spatial distribution of its material properties. The linearized elastodynamic equation may be written

$$\rho(x) \ddot{u} - \text{div}(\sigma) = f$$

$$\sigma = C(x) : \epsilon + \sigma_v$$

$$\epsilon = 1/2 [\text{grad}(u) + \text{grad}(u)^T]$$

Changing contour, curvature, wall thickness, fiber direction, network connectivity, joint placement, attached mass, surrounding fluid, or boundary conditions changes the governing operator. The visible form is therefore part of the mathematical specification of the physical response.

2.3 Distributed resonance

For free vibration, let $u(x,t) = \phi(x) \exp(i \omega t)$. After spatial discretization, the structural eigenproblem becomes

$$K \phi_n = \omega_n^2 M \phi_n$$

The mode belongs to the complete configuration: geometry, stiffness field, density field, attachments, prestress, and fluid loading. A frequency-domain fluid-loaded model can be represented as

$$[-\omega^2(M_s + M_f(\omega)) + i \omega(C_s + C_f(\omega)) + K_s] Q = F(\omega)$$

This is a reduced representation. In strongly nonlinear flapping or buckling regimes, added mass and damping need not remain linear or separable.

2.4 Energy balance and interpretive boundary

$$E_m = 1/2 \int_{\Omega} \rho |\dot{u}|^2 dV + 1/2 \int_{\Omega} \epsilon : C : \epsilon dV$$

$$dE_m/dt = P_{\text{active}} + P_{\text{boundary}} - P_{\text{diss}}$$

The sign of boundary power depends on the adopted traction convention; here positive boundary power denotes net work entering the structural domain.

The present study does not attribute the generation of energy to geometry itself. It examines how geometry and material architecture govern the pathways, impedances, modes, transformations, losses, and destinations through which available energy becomes dynamically expressed.

Part I - Flight: aeroelastic and sensory coupling

BEETLE FLIGHT: MATERIAL - FLOW - SENSORY COUPLING

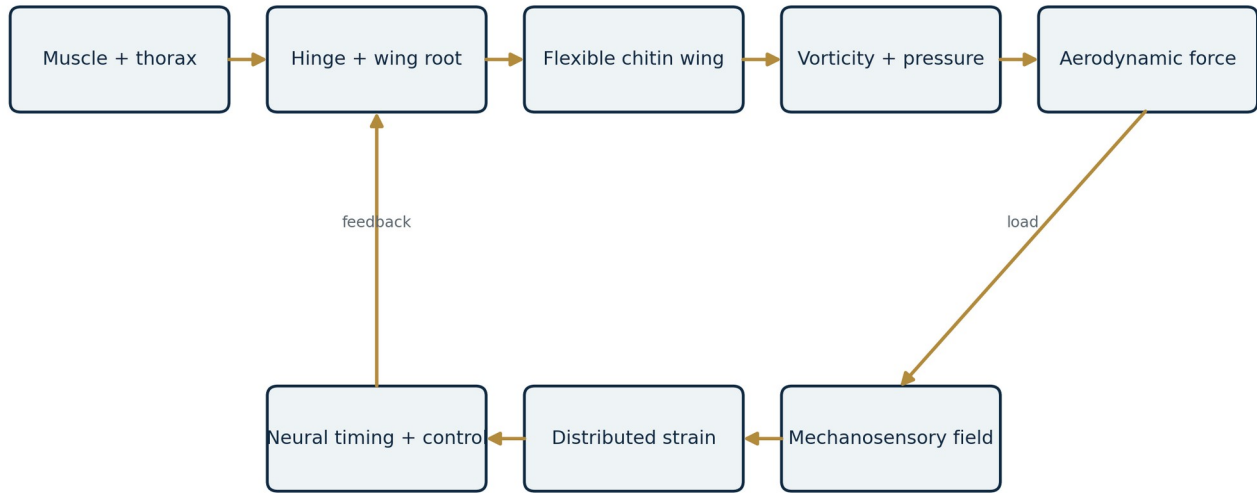


Figure 1. Established coupling loop for insect flight. The chitin surface participates as structure, fluid boundary, resonant medium, and strain-sensing substrate.

3. Scope note: phonon-related and exotic flight accounts

Accounts of beetle and insect flight sometimes extend beyond continuum aeroelasticity into claims involving phonons, specialized vibrational channels, microscopic lattice excitations, or other unusual wave-assisted lift mechanisms. These claims range from modest suggestions that multiscale vibration alters effective constitutive response to much stronger claims of a distinct lift contribution not represented by ordinary fluid-structure mechanics.

This paper neither affirms nor rejects such a mechanism. A direct treatment would require a defined microscopic mode or spectral band, a coupling law connecting that mode to wing-scale deformation or airflow, a measurable energy-transfer pathway, a prediction distinguishable from continuum aeroelasticity, and an experiment capable of falsification.

$$H_{\text{micro}} \rightarrow C^*(x, \omega) \rightarrow u(x, t) \rightarrow f_{\text{aero}}(t)$$

The necessary frequency-resolved constitutive measurements, multiscale coupling functions, and controlled aerodynamic data are not assembled here. The paper therefore confines its conclusions to measured material architecture and established continuum coupling.

4. Beetle flight as a coupled material-flow topology

The beetle flight apparatus includes thoracic deformation, muscle activation, wing-root geometry, articulating sclerites, veins, membranes, folding joints, local compliant regions, elytra, surrounding airflow, mechanosensory fields, and neural feedback. Its topology is not merely anatomical connectivity; it is the connectivity of force, displacement, fluid circulation, and information.

$$\mathbf{x}(t) = [q_s, \dot{q}_s, u_f, p_f, z_m, z_n]^T$$

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, \mathbf{u}_{\text{control}}, \mathbf{d})$$

Here $\mathbf{d}$ represents disturbances such as gusts, collisions, thermal shifts, or payload changes. The state description is schematic rather than a claim that all variables have been measured simultaneously in one beetle species.

5. Spatial stiffness, vein topology, and folding

Beetle hindwings combine thin membranes, reinforcing veins, folding lines, joints, and locally compliant regions. Resilin occurs at mechanically strategic sites associated with bending and repeated deformation. A simplified anisotropic plate model is

$$\rho_s(\mathbf{x}) h(\mathbf{x}) \ddot{\mathbf{w}} + \mathbf{D}[\mathbf{w}; \mathbf{D}_{ijkl}(\mathbf{x})] = \mathbf{p}_a(\mathbf{x}, t) + \mathbf{p}_i(\mathbf{x}, t)$$

For an isotropic plate the bending rigidity is

$$D = E h^3 / [12(1 - \nu^2)]$$

The cubic thickness dependence explains why modest thickness gradients can create large stiffness gradients. Real wings require orthotropic or anisotropic plate and beam descriptions, especially where veins and corrugations dominate.

The vein network may be represented as a weighted graph $G = (V, E)$, where nodes denote junctions or joints and edges carry bending rigidity, torsional rigidity, mass, and damping. This representation makes explicit how connectivity changes load redistribution, deformation paths, crack arrest, and lifecycle robustness.

6. Fluid-structure interaction and vorticity organization

$$u_f = \dot{u}_s \text{ on the interface}$$

$$\sigma_f n = \sigma_s n \text{ on the interface}$$

$$F_a = \int_S (-p_n + \tau_{fn}) dA$$

The wing changes the flow and the flow changes the wing. A generalized structural coordinate satisfies

$$m_n \ddot{q}_n + c_n \dot{q}_n + k_n q_n = Q_{m,n} + Q_{a,n}$$

$$Q_{a,n} = \int_S \phi_n \cdot \mathbf{t}_a dA$$

The mode shape determines how strongly a deformation couples to fluid traction. Modes producing useful twist, camber, or spanwise load redistribution can affect the flow more strongly than modes with poor geometric alignment.

The vorticity field $\boldsymbol{\omega} = \text{curl}(\mathbf{u})$ evolves, for an incompressible Newtonian fluid, according to

$$\partial_t \boldsymbol{\omega} + (\mathbf{u} \cdot \nabla) \boldsymbol{\omega} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \text{laplacian}(\boldsymbol{\omega})$$

The moving surface determines where vorticity forms, how coherent structures roll up, whether they remain attached, and how a wake interacts with later phases of the stroke. This provides the appropriate connection to topology: the operative object is the evolving relation among surface geometry, phase, deformation, and coherent flow structure.

7. Passive deformation as embodied control

A flexible wing does not require an independent control command at every material point. Distributed stiffness converts global root motion and aerodynamic pressure into local twist and camber.

$$\alpha(x,t) = \alpha_{\text{active}}(x,t) + \alpha_{\text{passive}}(x,t)$$

$$\alpha_{\text{passive}} = H_{\alpha}[p(x,t), \theta_r(t), C(x), h(x)]$$

This response can alter angle of attack, vortex attachment, rotational timing, and load distribution. Passive deformation is therefore not the absence of control; it is a control relation embodied in the material response, later modulated by active muscle and neural timing.

8. Resonant operation, damping, and lifecycle constraints

$$M \ddot{q} + C \dot{q} + K q = F_m(t) + F_a(t)$$

$$E_{\text{elastic}} = 1/2 \dot{q}^T K q$$

$$W_m = \text{closed_integral } \tau_m \cdot d\theta$$

Operating near a useful structural mode can reduce the active work required to reverse inertial motion. Yet maximal quality factor is not the biological objective. Maneuverability, disturbance rejection, phase control, and damage tolerance require finite damping and bandwidth.

$$\text{maximize } \int_0^{T_{\text{life}}} P_{\text{useful}}(t) dt$$

$$\text{subject to } D(T_{\text{life}}) < D_{\text{crit}} \text{ and } \sigma(x,t) < \sigma_{\text{allow}}(x,t)$$

9. Wing and hinge as sensory surfaces

Campaniform sensilla and other mechanoreceptors respond to cuticular strain and wing kinematics. A receptor output may be represented as

$$y_j(t) = S_j[\epsilon(x_j,t), \dot{\epsilon}(x_j,t), \theta(t)]$$

$$\epsilon(x_j,t) = \sum_n q_n(t) \epsilon_n(x_j)$$

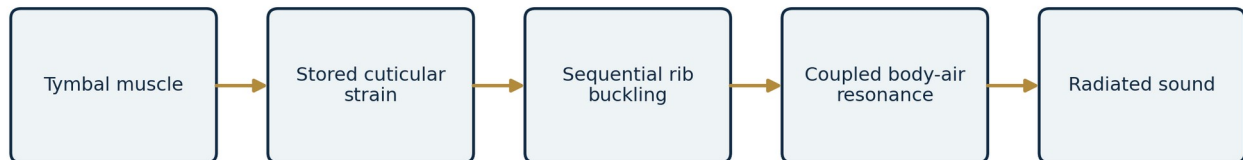
Sensor placement determines modal observability. A sensor near a modal node receives little signal from that mode; a sensor near a high-strain region receives a stronger projection. The surface therefore performs mechanical preprocessing before neural encoding.

10. Flight synthesis

The defensible claim is not that beetles possess an unexplained exotic force. It is that they embody a highly integrated, geometrically governed aeroelastic control system in which hierarchical material architecture produces distributed stiffness, inertia, damping, and sensing; these determine structural modes, passive deformation, vorticity organization, aerodynamic loading, and feedback.

Part II - Cicada acoustics

CICADA: NONLINEAR MECHANICAL-TO-ACOUSTIC TRANSDUCTION



Measured system-level transduction estimates approach tens of percent in a studied species.

Figure 2. The cicada energy path: muscle work is stored in the cuticle, released through sequential instability, filtered by coupled body-air resonance, and radiated as sound.

11. The cicada as an exceptionally efficient acoustic instrument

The cicada forms a compact mechanical-to-acoustic converter from differentiated cuticle, elastic tissue, muscle, and enclosed air. The central challenge is impedance transformation: a small solid source must transfer substantial mechanical work into low-impedance air.

$$Z_0 = \rho_a c$$

muscle shortening -> stored strain -> buckling -> resonance -> air excitation -> sound

12. Measured energetic conversion

For *Cyclochila australasiae*, published measurements report approximately 47.0 microjoules of tymbal-muscle work per contraction and approximately 45.1 microjoules of elastic energy released through the buckling sequence. The same study estimated mechanical-to-acoustic transduction efficiency at approximately 18-46%, depending on the measurement and accounting assumptions used by the authors.

The percentage must remain tied to the source authors' complete energetic accounting. It should not be recomputed from isolated headline values without matching pulse definition, bilateral source contributions, integration interval, and far-field radiation assumptions.

$$\eta_{ma} = E_{acoustic} / W_{mechanical}$$

The defensible conclusion is that the measured system transfers a substantial fraction, approaching tens of percent, of available mechanical work into radiated sound in a studied species.

13. Nonlinear sequential buckling

Let q_j describe rib j . A reduced coupled model is

$$m_j \ddot{q}_j + c_j \dot{q}_j + \partial U / \partial q_j + \sum_{k \neq j} \kappa_{jk} (q_j - q_k) = F_j(t)$$

The potential U contains unstable regions associated with shell buckling. One comparatively slow muscle contraction can therefore trigger several rapid structural transitions and a train of high-frequency impulses.

14. Configuration-dependent resonance

Measurements in a well-studied cicada tymbal show resonance shifting from roughly 5.5 kHz toward approximately 4.3 kHz as the structure enters the inward-buckled configuration. The tangent stiffness changes with configuration:

$$K_T(q) = \partial^2 U / \partial q^2$$

$$K_T(q) \phi_n = \omega_n(q)^2 M \phi_n$$

The process that releases strain energy also changes the modal conditions through which the released energy rings.

15. Acoustic loading, radiation, and intense local pressure

$$Z_a(\omega) = R_a(\omega) + i X_a(\omega)$$

$$P_{\text{rad}} = 1/2 R_a |v_n|^2$$

The resistive component carries average power into outgoing waves; the reactive component stores and returns energy near the source. Efficient radiation therefore requires both resonant response and useful radiation resistance.

Reported local transient pressures inside the abdominal air system can reach extreme sound-pressure levels. Sound-pressure level is

$$L_p = 20 \log_{10}(p_{\text{rms}} / p_{\text{ref}}), \quad p_{\text{ref}} = 20 \text{ microPa}$$

For illustration, 142 dB SPL corresponds to approximately 252 Pa rms. These internal or near-field transient values must not be confused with ordinary far-field exposure levels or used alone as an efficiency measure.

16. Coupled structural-acoustic resonance

A simple Helmholtz approximation uses

$$\omega_H = 1 / \sqrt{M_a C_a}$$

$$M_a = \rho_a L_{\text{eff}} / A_n, \quad C_a = V / (\rho_a c^2)$$

A real cicada body is more complex because tymbal, abdominal air, tympana, openings, and flexible walls all participate. A reduced coupled eigenproblem is

$$[K_s - \omega^2 M_s, C_{sa}; C_{as}, K_a - \omega^2 M_a] [\phi_s; \phi_a] = 0$$

The 2025 metastructure treatment further supports a model of sequential bistable elements coupled to resonant filtering structures. It should be cited as a recent modeling framework, not treated as proof that every cicada species shares one identical band-gap architecture.

17. Efficiency hierarchy

The paper distinguishes elastic storage and release, mechanical-to-acoustic transduction, metabolic-to-acoustic conversion, and far-field radiation efficiency. These quantities have different denominators and experimental requirements.

$$\eta_e = E_{\text{released}} / W_{\text{muscle}}$$

$$\eta_{\text{meta-ac}} = \eta_{\text{muscle}} \eta_{\text{ma}}$$

The near equality of reported muscle work and released tymbal energy indicates close energetic matching in the measured specimens, but uncertainty prevents treating their quotient as a universal exact efficiency. The 18-46% transduction estimate remains the strongest system-level figure and is retained with its source-specific conditions.

18. Cicada synthesis

The cicada cuticle serves simultaneously as frame, elastic store, nonlinear switching structure, resonator, pressure boundary, impedance transformer, radiating surface, and fatigue-bearing anatomy. Its extraordinary loudness becomes an engineering result because the measured energy accounting indicates direct transfer of a large fraction of mechanical work into sound, not merely a high local pressure amplitude.

Part III - Honeybee thermal regulation

HONEYBEE COLONY: DISTRIBUTED THERMAL FIELD CONTROL

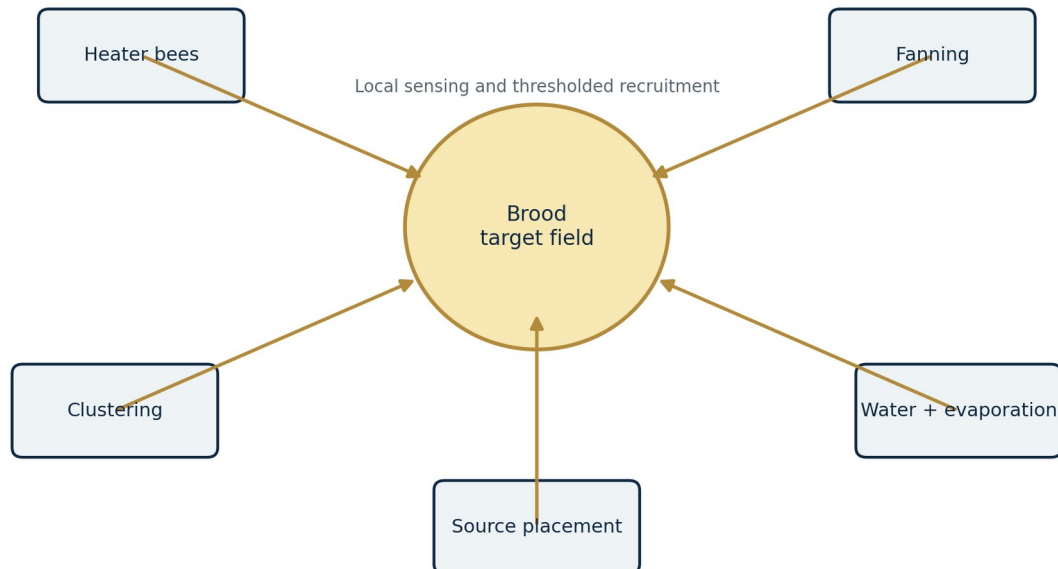


Figure 3. Colony thermoregulation as distributed field control. Workers alter heat generation, source placement, insulation, airflow, and evaporation in response to local conditions.

19. A collective thermal-control system

The honeybee colony converts chemical energy into a spatially shaped and actively regulated thermal field. The achievement is not merely heat production, but selective placement, retention, transport, withdrawal, and redirection of heat for brood development, winter survival, cooling, and defense.

flight-muscle metabolism → thoracic heat → contact → comb conduction → colony field

Brood-nest temperature is commonly maintained in a narrow range near 33-36 C, while peripheral regions may vary more strongly. The colony therefore uses selective thermal zoning rather than brute-force whole-volume heating.

20. Distinct thermal efficiencies

$$\begin{aligned} P_{\text{heat}} &= \eta_{\text{met}} \rightarrow \text{heat } P_{\text{chem}} \\ \eta_{\text{delivery}} &= Q_{\text{brood}} / Q_{\text{generated}} \\ \eta_{\text{retention}} &= 1 - Q_{\text{loss}} / Q_{\text{input}} \end{aligned}$$

A single universal colony thermal-efficiency percentage is not presently justified because a closed energy budget would require complete measurements of food consumption, metabolic heat, storage, conduction, convection, evaporation, ventilation, and boundary losses. The paper therefore emphasizes regulatory effectiveness, source-placement efficiency, and retention efficiency rather than inventing one number.

21. Heater bees as movable thermal actuators

$$\begin{aligned} P_{i,\text{useful}} &= \eta_i(x_i, A_{c,i}, R_{th,i}, T_i - T_b) P_i \\ Q_{i,\text{dot}} &= k_{\text{eff}} A_{c,i} (T_i - T_b) / L_i \end{aligned}$$

Workers can increase useful delivery without proportionally increasing heat generation by moving closer to brood, occupying empty cells, increasing contact area, and shortening the conduction path. The same generated heat has different biological value in different geometries.

22. The cluster as an active porous material

$$\rho_{\text{eff}} c_{\text{eff}} \partial_t T = \text{div}(k_{\text{eff}} \text{grad } T) - \rho_a c_a v \cdot \text{grad } T + q_{m,\text{dot}}$$

Unlike inert insulation, the cluster can alter effective conductivity, porosity, metabolic source density, and airflow through behavior. This allows the colony to reduce losses as well as increase production.

23. Cooling and distributed control

$$\begin{aligned} C_h dT/dt &= P_{\text{met}} - Q_{\text{cond}} - Q_{\text{conv}} - Q_{\text{rad}} - Q_{\text{evap}} \\ Q_{\text{conv}} &= h A (T - T_a) \\ Q_{\text{evap}} &= m_w \dot{h}_{fg} \end{aligned}$$

Fanning modifies airflow and convective transfer; transported water enables latent-heat removal. Individual workers respond to local conditions through heating, fanning, water handling, clustering, or relocation.

$$u_i(t) = R_i[T(x_i,t), \text{grad } T(x_i,t), \text{brood cues, age, internal state}]$$

A distribution of response thresholds can produce graded recruitment rather than synchronized all-or-none correction. This supplies a mathematically plausible mechanism for stable colony-level behavior without a central thermostat.

24. Winter zoning and regulatory efficiency

$$\Omega_{\text{hive}} = \Omega_{\text{core}} \cup \Omega_{\text{mantle}} \cup \Omega_{\text{periphery}}$$

$$T_{\text{core}} > T_{\text{mantle}} > T_{\text{periphery}}$$

The colony permits noncritical regions to cool while defending the biologically necessary core. Efficiency therefore emerges from selective regulation, dynamic insulation, and spatial allocation of fuel use.

25. Defensive bee-balling

Japanese honeybees and related Asian lineages can surround giant hornets in a dense defensive ball. Hundreds of workers activate thoracic musculature and maintain a confined microenvironment whose core can approach approximately 46 C.

$$C_{\text{ball}} dT/dt = N_b P_b - U A (T - T_a) - m_{\text{air}} c_p (T - T_a)$$

Temperature contributes strongly, but experiments indicate that elevated carbon dioxide also contributes materially; mechanical restraint and, in some encounters, stinging may contribute as well.

$$\text{Lethality} = L(T, p_{\text{CO}_2}, p_{\text{O}_2}, \text{exposure time}, \text{restraint}, \text{stinging})$$

The defensive ball should therefore be described as a collectively generated thermal, respiratory, and mechanical microenvironment rather than as heat alone.

26. Bee synthesis

$$\eta_{\text{colony}} = \eta_{\text{conversion}} \eta_{\text{placement}} \eta_{\text{retention}} \eta_{\text{recruitment}} \eta_{\text{control}}$$

This product is conceptual rather than a measured universal identity. It highlights that colony performance depends on several linked efficiencies: converting chemical energy, placing sources, retaining heat, recruiting workers proportionally, and avoiding under- or over-correction.

$$\text{individual anatomy} + \text{distributed rules} + \text{collective geometry} = \text{regulated thermal field}$$

Part IV - Unified interpretation

27. Three expressions of one architectural principle

$$\text{Beetle: } E_{\text{chem}} \rightarrow E_{\text{elastic/kinetic}} \rightarrow E_{\text{fluid}}$$

$$\text{Cicada: } E_{\text{chem}} \rightarrow E_{\text{elastic}} \rightarrow E_{\text{acoustic}}$$

$$\text{Honeybee: } E_{\text{chem}} \rightarrow E_{\text{thermal}} \rightarrow E_{\text{colony field}}$$

Let E collect chemical, elastic, kinetic, fluid, acoustic, and thermal energy accounts. A bookkeeping representation is

$$E_{\text{dot}} = T(x,t) E + S - D$$

This transfer-matrix notation is a systems abstraction, not a fundamental linear law for all regimes. The present analysis makes no claim that evolutionary geometry originates the available energy or alters governing conservation relations. It examines how developmentally expressed geometry and material coefficients modify coupling among structural, fluid, acoustic, sensory, and thermal domains.

28. Resonance as meaningful relation

Resonance here describes relations among forcing, natural modes, damping, fluid or acoustic loading, sensory timing, fatigue limits, and operating duration. The biological objective is not maximum resonance amplitude in isolation, but useful lifetime performance under control and damage constraints.

$$D(t) = \int_0^t \Phi(\sigma_a, \epsilon_a, \omega, T, \text{hydration}) d\tau$$

$$D(T_{\text{life}}) < D_{\text{crit}}$$

29. Developmental geometry and evolutionary filtering

$$D_{\text{map}}: g \rightarrow \{\Omega, C(x), \rho(x), \eta(x), k_t(x)\}$$

$$P_{\text{map}}: \text{phenotype} \rightarrow \{\text{flight, sound, heat, damage, stability}\}$$

Selection acts on integrated phenotypes produced by developmental rules. One geometric change can alter resonance, flow coupling, sensing, thermal transfer, damage tolerance, and energetic cost simultaneously. The observed form therefore occupies a historically constrained Pareto region rather than proving a global optimum.

30. Evidence tiers

Tier	Status and examples
Tier I - Directly established	Spatially differentiated insect cuticle; vein and membrane effects on wing mechanics; aeroelastic deformation; mechanosensory encoding; tymbal buckling and resonance; source-specific cicada transduction estimates; brood heating, clustering, ventilation, evaporation, and hot defensive bee balls.
Tier II - Consequences of established physics	Geometry and constitutive fields alter modes; modes alter fluid and acoustic coupling; receptor placement alters observability; source placement and cluster geometry alter thermal delivery and retention; lifecycle constraints bound useful resonance.
Tier III - Open hypotheses	Specialized elastic-wave filtering in flight; microscopic phonon effects not represented by continuum parameters; useful reactive work from internal air volumes; more precise multiscale co-tuning than current measurements resolve.
Excluded as established fact	A distinct phonon-generated lift mechanism; universal resonant optimization of all cuticle; one Helmholtz model for all cicadas; one universal numerical efficiency for honeybee thermoregulation.

Conclusion

Flying beetles, singing cicadas, and thermoregulating honeybees represent three expressions of a broader arthropod construction principle. In beetles, differentiated chitin surfaces become folding structures, resonant transmission systems, aerodynamic boundaries, and sensory fields. In cicadas, a ribbed cuticular shell and body-air system become an unusually efficient nonlinear acoustic instrument. In honeybees, individual thoracic heat production becomes part of a colony-scale field-control system in which worker placement, clustering, ventilation, evaporation, and collective geometry regulate conditions far larger than any individual.

The deepest defensible conclusion is that biochemistry expresses geometrically organized living materials whose resonant, aerodynamic, acoustic, sensory, and thermal behavior remains coordinated with the operating requirements and survivable life of the organism.

Its material geometry is the continuously operating technology.

Mathematical audit notes

All displayed equations are either governing relations, reduced models, scaling laws, or systems abstractions. They are labeled in the prose according to their status.

The plate-rigidity equation applies only to a homogeneous isotropic thin plate and is used solely to show cubic thickness scaling. Real insect wings require anisotropic, heterogeneous, and often geometrically nonlinear models.

The linear modal equations describe local or reduced behavior. Flapping wings, buckling tymbals, and active bee clusters can be strongly nonlinear and time varying.

The cicada 18-46% estimate remains source-specific. The paper does not recompute it from unmatched headline quantities.

The honeybee multiplicative efficiency expression is conceptual and does not claim measured independence among factors.

The transfer-matrix energy representation is bookkeeping language, not a claim of universal linear energy dynamics.

The paper avoids attributing energy creation to geometry and confines itself to energy routing, coupling, transformation, storage, radiation, and loss.

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